



Econ 139: Labor Economics

Final Review

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Dec 3 — Write down your first and last name



There are generally three types of questions: (1) calculation based questions (i.e. given production function and labor supply curve, find equilibrium wages/employment), (2) questions about conceptual frameworks (i.e. explain the concept of income vs. substitution effects) or (3) questions about papers we have studied. Which would you most like to focus on during the next review session



Which topics would you like to focus on next class. Pick all that apply



Overview

- High level review/tips
- Go through some questions from practice final

Theoretical Models

- We have discussed a number of theoretical frameworks
 - Perfectly Competitive Models
 - Monopsonistic models
 - Task-based approach to production
 - Labor supply models

Empirics

- We have discussed empirics in many sections
- **Trends:** wage inequality, labor force participation, etc.
 - Key here is to understand exactly what you are measuring, what it does and does not capture
 - Know the main facts about the “big questions in labor economics”
- **Causal Effects:** What is the effect of X on some outcome Y of interest
 - Key here is to understand how we go from correlation to causation
 - Easiest solution: run an experiment, like we saw with the negative income tax experiments in the Labor Supply section
 - More complex: Use observation variation across people, places etc, but argue variation is not driven by other factors (i.e. regional exposure to trade, regional exposure to robots, natural experiments)



Imagine a single firm in the economy sees an increase in their efficiency which raises the marginal revenue product of each worker. In a perfectly competitive market, how does this change the wage of the workers they employ



Imagine every firm in the economy sees an increase in their efficiency which raises the marginal revenue product of each worker. In a perfectly competitive market, how does this change the wage of the workers?

Practice Final 1 Question 2/ Monopsony Review

Consider the table below that displays both the output for a number of workers, as well as the wage you must pay to attract that number of workers. You can therefore assume this firm is a monopsonist in the labor market.

Workers (L)	Output (Q)	Wage (W)
1	5	\$10
2	10	\$15
3	15	\$20
4	20	\$25

1. Compute the marginal revenue product of labor and the marginal factor cost for each worker. You can assume the output sells for 4 dollars per unit.

Further Monopsony/Perfect Competition Review

- Final 2 Problem 2 is a great monopsony review
 - Know intuitively how the two models (perfectly competitive vs monopsony) differ
 - Also understand how to draw the relevant curves
 - i.e. for monopsony need labor supply curve, marginal revenue product of labor curve, and marginal factor cost to find equilibrium employment
 - For competitive need the market wage and marginal revenue product of labor curve

Task-Based framework of Production

- Autor, Levy and Murnane (2003) conceptualize production as a bundle of tasks
 - Break down tasks into cognitive/manual vs. routine/non-routine
- Why is this useful?

Task-Based framework of Production

- Autor, Levy and Murnane (2003) conceptualize production as a bundle of tasks
 - Break down tasks into cognitive/manual vs. routine/non-routine
- Why is this useful?
 - Rise of personal computers in the 1980s corresponded to replacement of routine tasks that were programmable. This “hollowed out” the middle of the income distribution

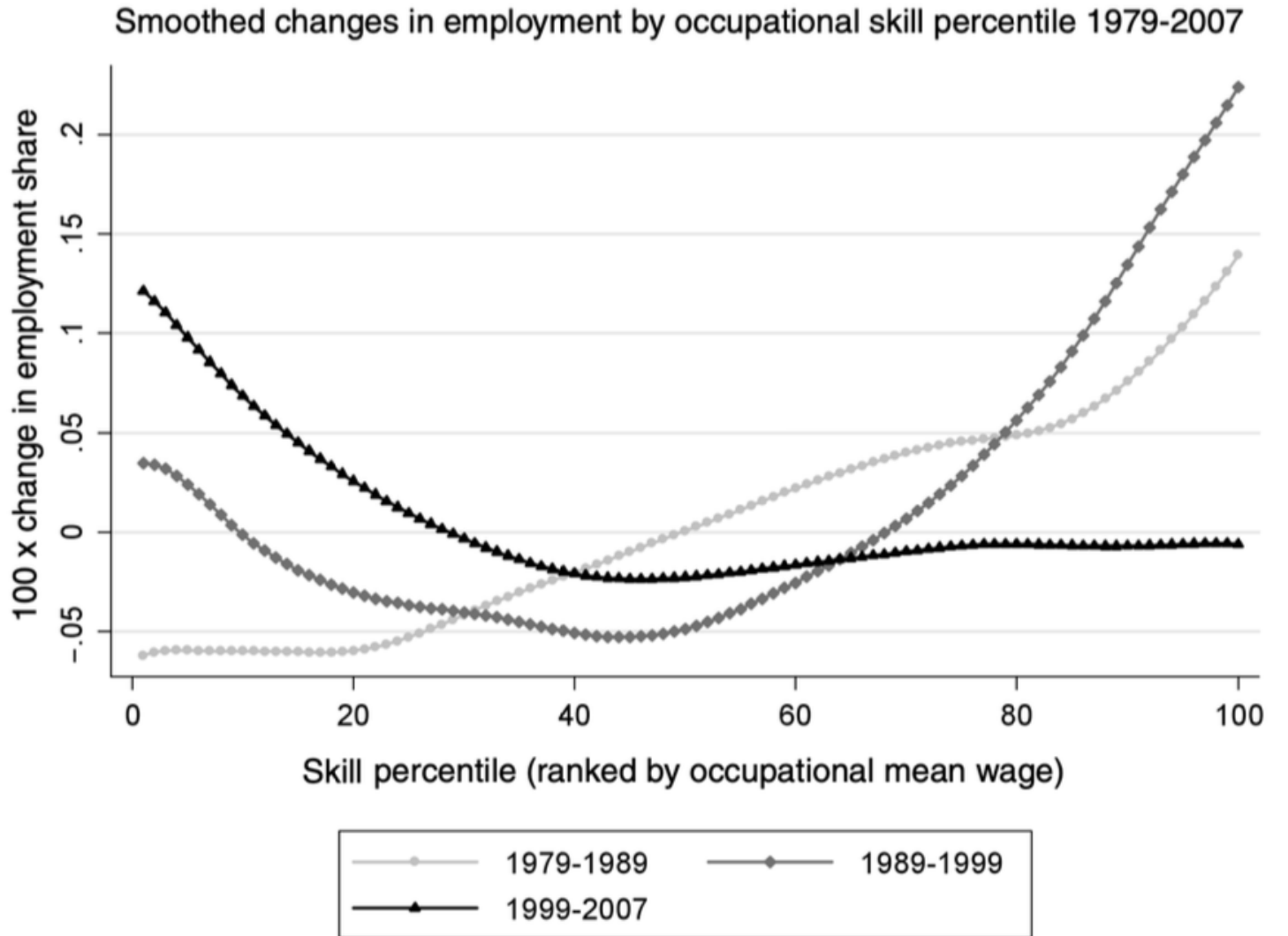


In the 1980s, the number of college graduates continued to rise. If nothing else changed, I would expect...



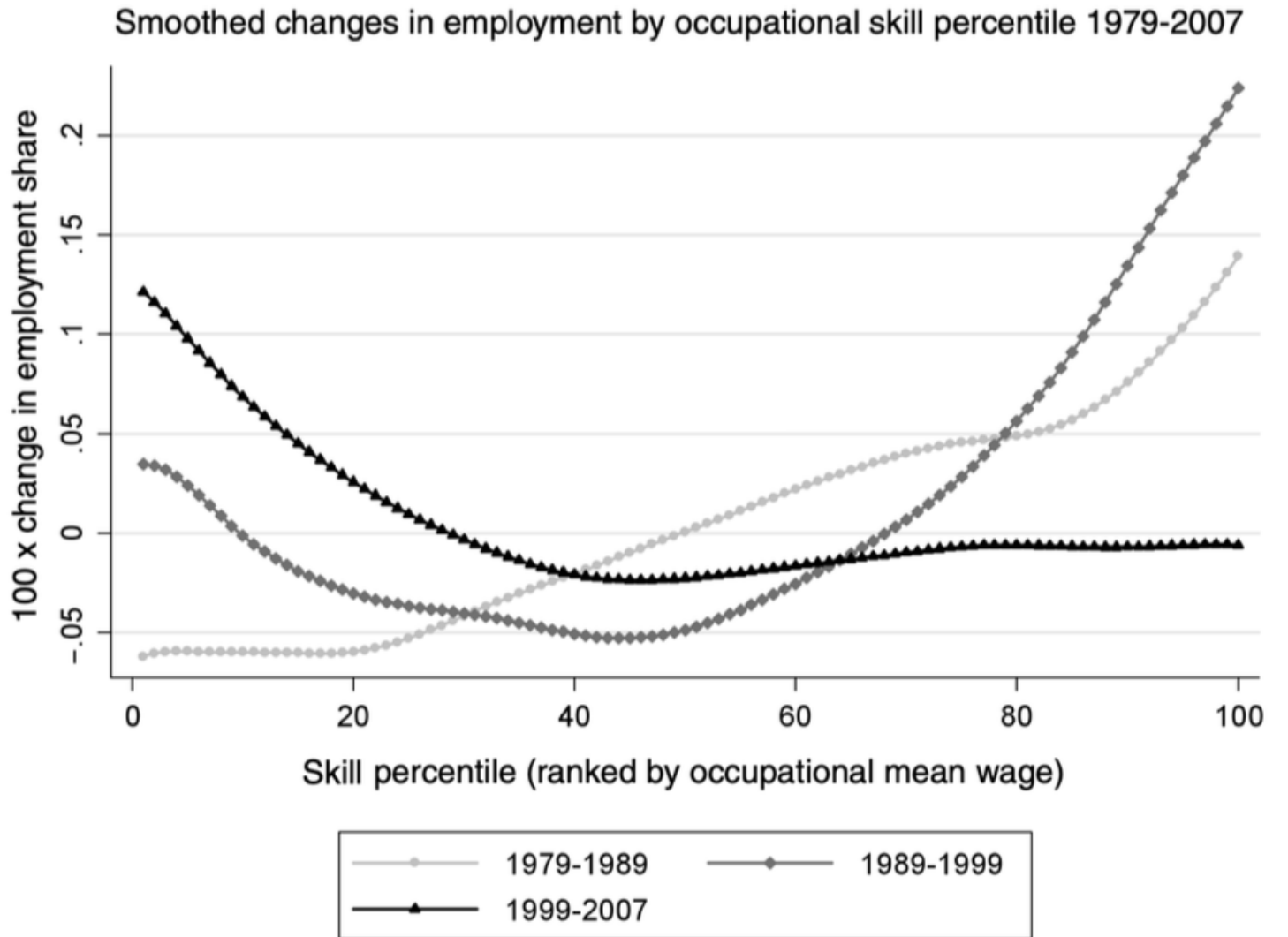
Can the rise of computers potentially explain why inequality rose during the 1980s despite rising supply of college graduates. Explain your answer

Task-Based framework of Production



Task-Based framework of Production

1. Explain the concept of job polarization?
2. What drove job polarization?



A Calculation-Based Question

Imagine a firm needs to fill three tasks to produce one unit of output. For example, the tasks could be: (1) assembling the product, (2) managing inventory and (3) designing the product/overseeing production processes. The firm can hire two types of workers: High-Skill (HS) workers who can perform all three tasks, and Low-Skill (LS) workers who can perform tasks (1) and (2). Below we present a table that shows the cost of hiring each type of worker per task. The cost is different for different tasks because it may take more time for a worker to perform a certain task relative to another task.

Worker Type	Task 1 Cost	Task 2 Cost	Task 3 Cost
High-Skill (HS)	\$20	\$40	\$40
Low-Skill (LS)	\$10	\$20	—



Dec 5 — Write down your first and last name

A Calculation-Based Question

Worker Type	Task 1 Cost	Task 2 Cost	Task 3 Cost
High-Skill (HS)	\$20	\$40	\$40
Low-Skill (LS)	\$10	\$20	—

1. What is total cost of producing one unit of output?

A Calculation-Based Question

Worker Type	Task 1 Cost	Task 2 Cost	Task 3 Cost
High-Skill (HS)	\$20	\$40	\$40
Low-Skill (LS)	\$10	\$20	—

2. What if increase in college attendance shifted labor supply of college graduates so that high-skill workers now cost less (15, 30, 30). How does this change output decision of firm?

A Calculation-Based Question

Input Type	Task 1 Cost	Task 2 Cost	Task 3 Cost
High-Skill (HS)	\$20	\$40	\$40
Low-Skill (LS)	\$10	\$20	—
Machine	\$9	—	—

3. How does adopting machine impact demand for high-skill and low-skill workers. Why does the answer it depend on the price of the machine?

Labor Supply

Imagine a worker can work up to 15 hours per day. The worker earns a wage of 10 dollars per hour worked.

1. Draw the worker's budget constraint with hours of leisure on the x-axis and income on the y-axis. Label the axes and the intercepts.

Labor Supply

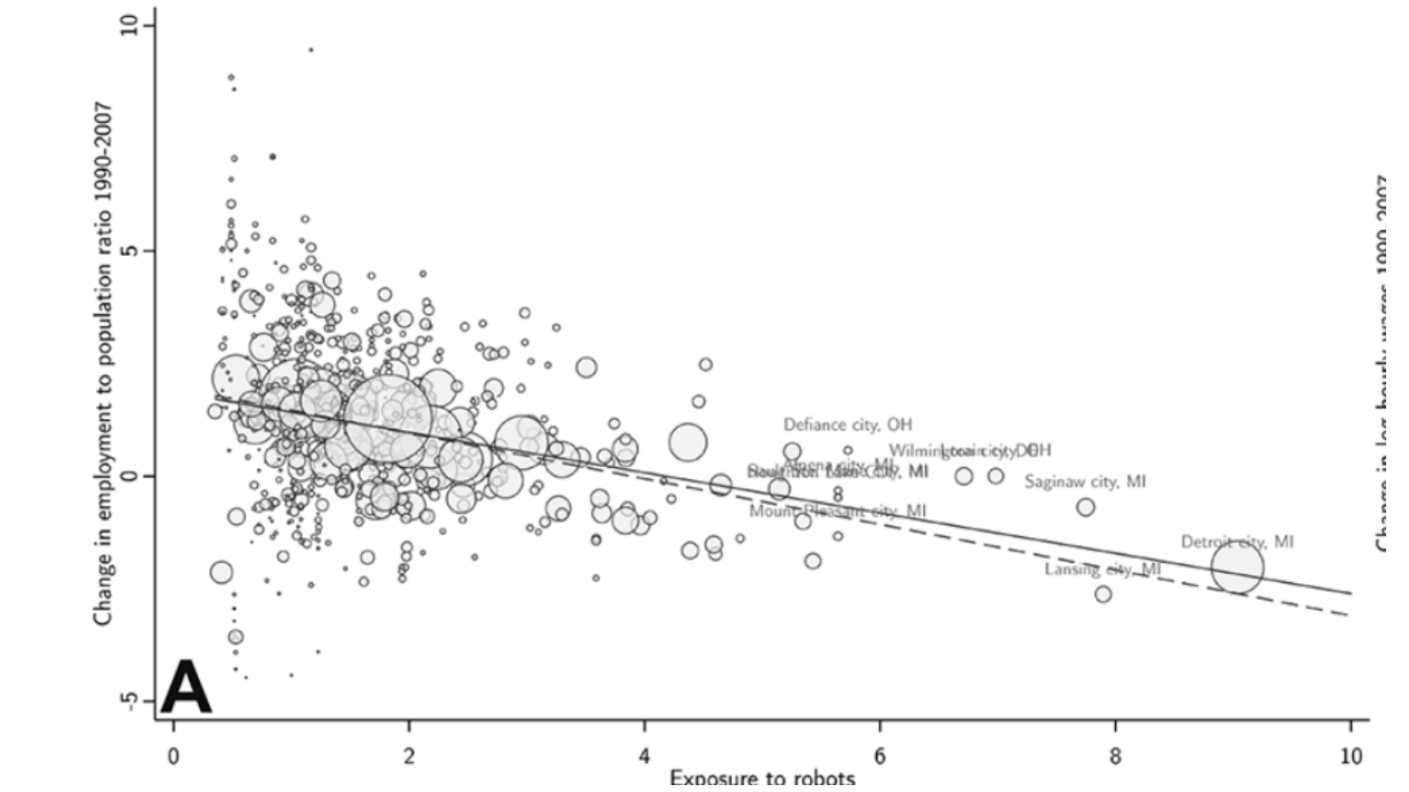
2. Imagine the worker becomes eligible for a work incentive program that adds a bonus of 5 dollars per hour for the first 5 hours worked. Draw the new budget constraint on the same graph as in part (a). Label the axes and the intercepts.
3. Draw an indifference curve that is valid for a person who optimally chooses to work 10 hours per day before the work incentive program. For an individual that was planning to work 10 hours per day before the work incentive program, explain how the program would affect their labor supply?
4. For a person planning to work zero hours per day before the work incentive program, explain how the program would affect their labor supply? You do not need to draw an indifference curve for this part.

Automation and Exposure Across Commuting Zones

- We studied a couple of different papers that took the commuting zone of the worker as the labor market
 - Azar, Marinescu, and Steinbaum (2022) – concentration
 - Acemoglu and Restrepo (2020) – exposure to automation
 - Autor, Dorn and Hanson (2012) – exposure to trade
- These papers all share some similarities in the approach
 - We need to be careful about exactly what we are measuring
 - We need to be careful that our measure of interest is not correlated with other factors that vary across regions

Automation

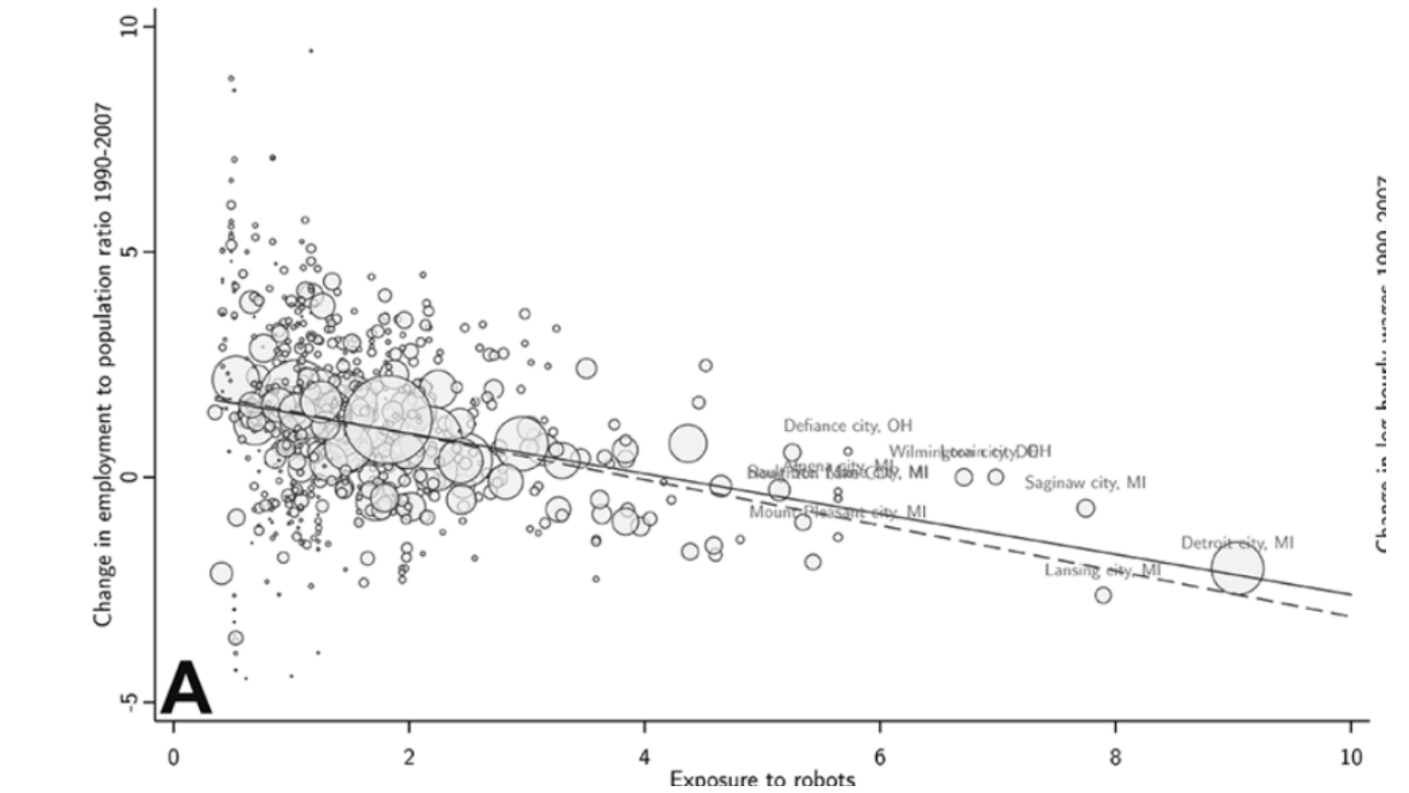
- Acemoglu and Restrepo (2020)



Impact of Exposure to Robots on Employment to Population Ratio

Automation

- Acemoglu and Restrepo (2020)



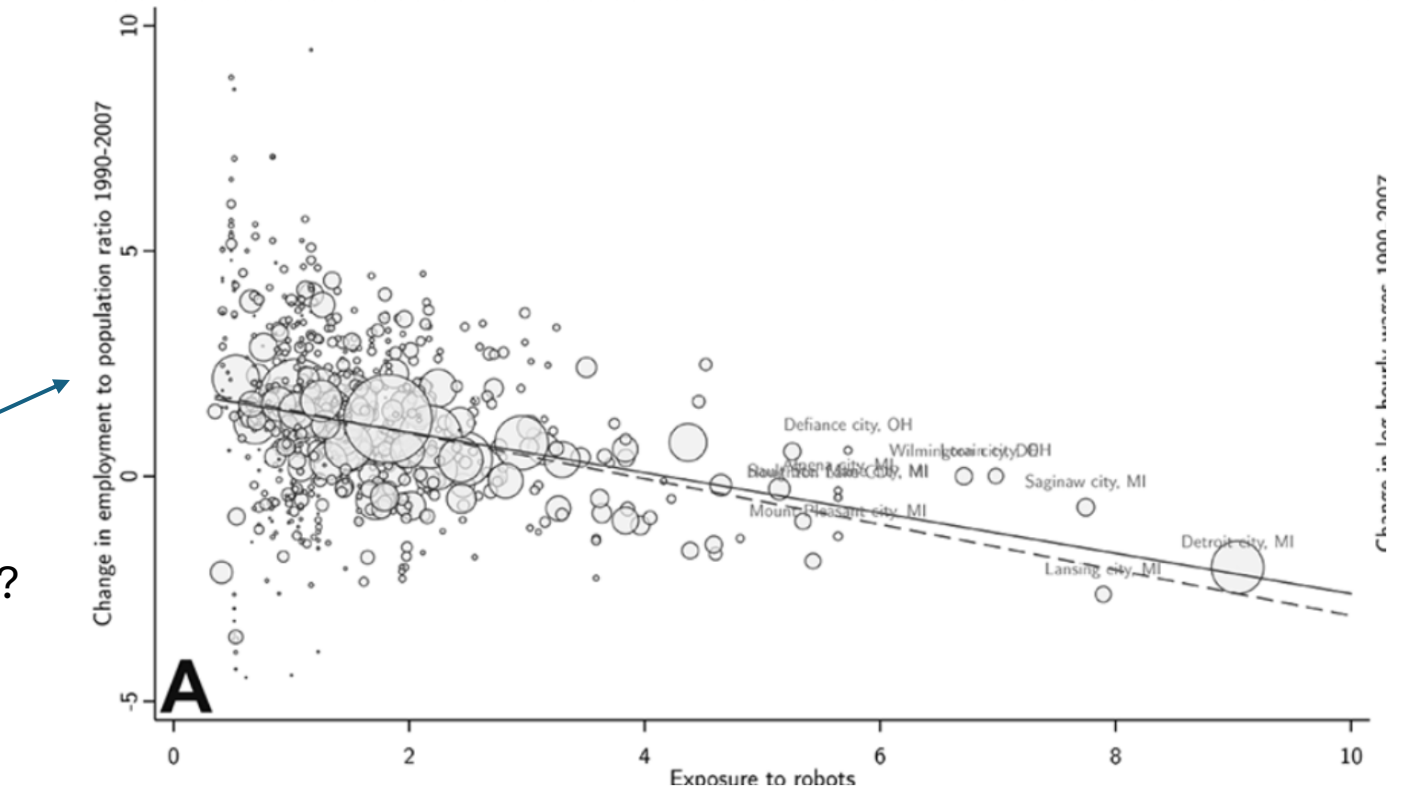
Impact of Exposure to Robots on Employment to Population Ratio

Number of additional robots per 1000 workers in the commuting zone

Automation

- Acemoglu and Restrepo (2020)

Change in Employment to population ratio?



Number of additional robots per 1000 workers in the commuting zone



What is one potential confounding factor in this research design?

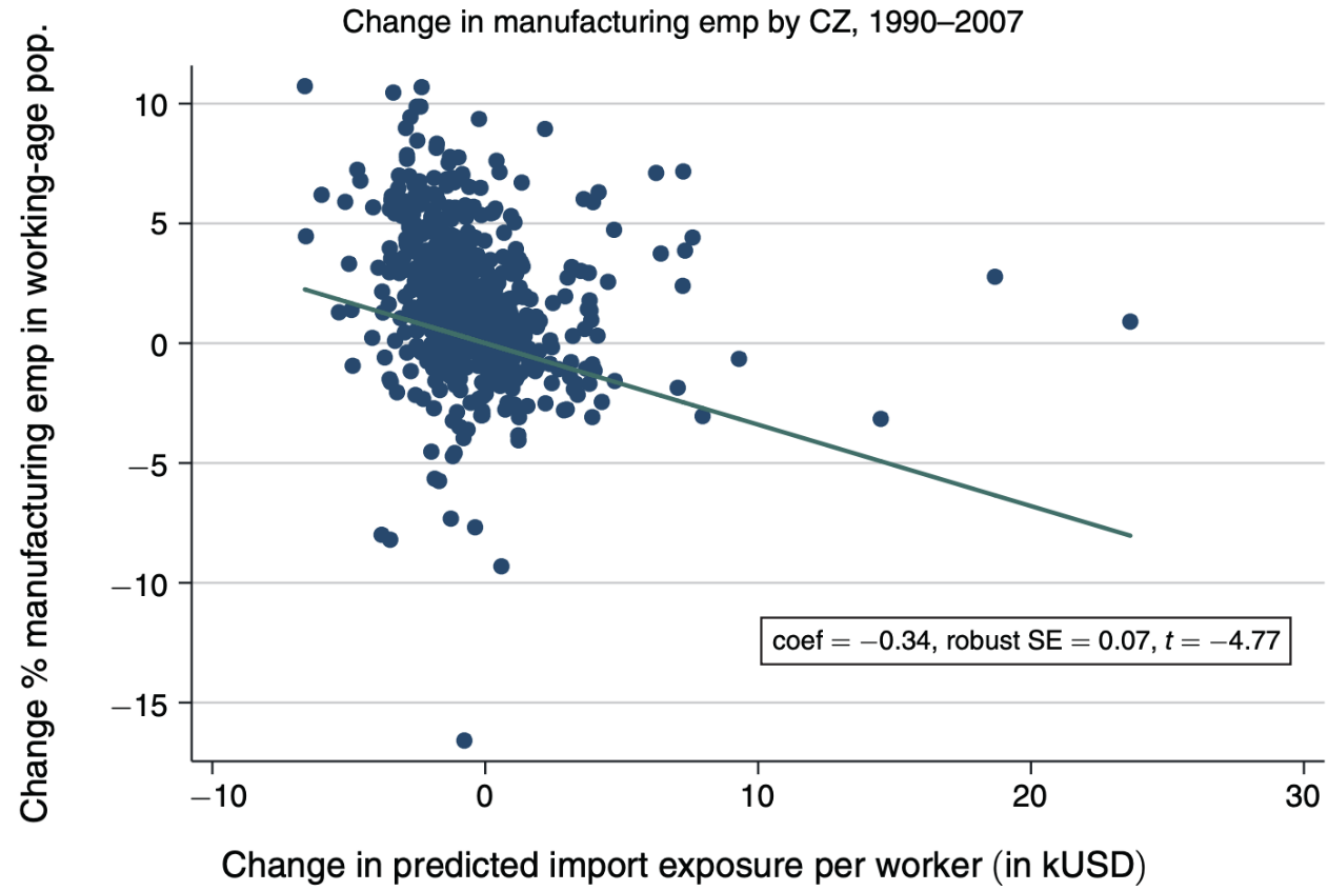


How would the confounding factor you chose bias the results?

Trade

- Autor, Dorn and Hanson (2013)

Panel B. OLS reduced form regression, full sample



Big Trends in Labor Economics

- Make sure to review the big trends we studied in the first week of the class
- How are these trends measured
- What are some drivers of these trends (i.e. what did we find could explain increases in wage inequality)

